

Anesthesia for Renal Diseases and Electrolyte Imbalances MCQ – Chapter 22

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 22)

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. What is the cause of anemia in patients with renal impairment?

- A. Iron deficiency
- B. Decreased erythropoietin or decreased intake due to anorexia
- C. Vitamin B12 deficiency
- D. Folate deficiency

Ans: B

Q2. What percentage of patients with chronic renal disease exhibit hypertension?

- A. More than 50%
- B. More than 60%
- C. More than 70%
- D. More than 80%

Ans: D

Q3. What is the most common endocrine cause of chronic kidney disease?

- A. Hypothyroidism
- B. Cushing's syndrome
- C. Diabetes
- D. Hyperparathyroidism

Ans: C

Q4. Patients with acute kidney injury (AKI) should not undergo elective surgery until:

- A. Hemoglobin normalizes
- B. The injury is fully resolved
- C. Blood pressure is controlled
- D. Potassium is less than 4 mEq/L

Ans: B

Q5. What should be the target blood pressure before surgery in CKD patients?

- A. < 120/80 mm Hg
- B. < 130/80 mm Hg
- C. < 140/90 mm Hg
- D. < 150/90 mm Hg

Ans: B

Q6. How many hours before surgery should dialysis be performed in CKD patients?

- A. 12 hours
- B. 24 hours
- C. 48 hours
- D. 6 hours

Ans: B

Q7. Why should premedication with benzodiazepines be avoided in renal patients?

- A. They are nephrotoxic
- B. They cause hyperkalemia
- C. CNS of these patients is extremely sensitive to sedatives and narcotics
- D. They increase bleeding tendency

Ans: C

Q8. What is the minimum urine output that should be maintained intraoperatively in renal patients? A. > 0.3 mL/kg/hr B. > 0.5 mL/kg/hr C. > 1.0 mL/kg/hr D. > 2.0 mL/kg/hr	Ans: B
Q9. Which selective dopamine-1 agonist is preferred over dopamine for maintaining urine output in major surgeries on renal patients? A. Dobutamine B. Fenoldopam C. Milrinone D. Noradrenaline	Ans: B
Q10. Why should radial and ulnar artery be spared in renal patients requiring invasive blood pressure monitoring? A. They have high infection risk B. They may be required for hemodialysis in future C. They are calcified in renal patients D. They give inaccurate readings	Ans: B
Q11. Which artery is most often chosen for invasive BP monitoring in renal patients? A. Radial artery B. Ulnar artery C. Dorsalis pedis D. Brachial artery	Ans: C
Q12. BP cuff should not be applied in the limb having: A. Peripheral neuropathy B. AV fistula C. Edema D. Cellulitis	Ans: B
Q13. Why should central neuraxial blocks be avoided in renal patients? A. Risk of epidural abscess B. Associated uremic coagulopathy, uremic polyneuropathy, and intolerance to extra fluids C. High failure rate D. Risk of nephrotoxicity from local anesthetics	Ans: B
Q14. What is the induction agent of choice in renal patients? A. Thiopentone B. Ketamine C. Propofol D. Etomidate	Ans: C

Q15. Why should propofol be used in small doses in renal patients? A. It is nephrotoxic in high doses B. It causes hyperkalemia C. Exaggerated response of CNS due to disrupted blood brain barrier in uremia D. It causes metabolic acidosis	Ans: C
Q16. What is the inhalational agent of choice for maintenance in renal patients? A. Sevoflurane B. Isoflurane C. Desflurane D. Halothane	Ans: C
Q17. Why should sevoflurane not be used in renal patients? A. It causes bronchospasm B. It produces high levels of fluoride approaching renal threshold of 50 µmol/L C. It is hepatotoxic D. It causes malignant hyperthermia	Ans: B
Q18. Why should succinylcholine not be used in renal patients? A. It causes malignant hyperthermia B. It is metabolized by kidneys C. These patients exhibit hyperkalemia D. It causes prolonged block	Ans: C
Q19. Which non-depolarizing muscle relaxants are the agents of choice in renal patients? A. Pancuronium and Vecuronium B. Atracurium and Cis-atracurium C. Rocuronium and Vecuronium D. Mivacurium and Doxacurium	Ans: B
Q20. What is the opioid of choice in renal patients? A. Morphine B. Pethidine C. Remifentanil D. Codeine	Ans: C
Q21. Which long-acting opioids are NOT recommended in renal patients? A. Fentanyl and Alfentanil B. Morphine and Pethidine C. Remifentanil and Sufentanil D. Alfentanil and Sufentanil	Ans: B

Q22. Why is the traditional concept of avoiding ringer lactate in renal patients no longer valid? A. Ringer lactate is cheaper B. Normal saline causes hyperchloremic metabolic acidosis producing more hyperkalemia C. Ringer lactate has less sodium D. Normal saline causes hyponatremia	Ans: B
Q23. Which fluid is preferred in renal patients for intraoperative use? A. Normal saline (0.9%) B. Ringer lactate C. 5% Dextrose D. Hetastarch	Ans: B
Q24. If ringer lactate cannot be used, which alternative fluid is preferred in renal patients? A. Normal saline 0.9% B. 0.45% saline or 0.18% saline with dextrose C. Hetastarch 6% D. 5% albumin	Ans: B
Q25. Why should colloids (hydroxyethyl starch) be avoided in renal patients? A. They cause anaphylaxis B. They have been found to exacerbate renal injury C. They cause hypernatremia D. They are expensive	Ans: B
Q26. If a colloid must be used in renal patients, which one is preferred? A. Hetastarch B. Dextran C. Gelatins (Gelofusine) D. Albumin 20%	Ans: C
Q27. What molecular weight colloids must NOT be used in renal patients? A. > 100 kDa B. > 150 kDa C. > 200 kDa D. > 250 kDa	Ans: C
Q28. Which drugs should be avoided postoperatively in renal patients for pain management? A. Paracetamol B. Non-steroidal anti-inflammatory drugs (NSAIDs) C. Tramadol D. Fentanyl	Ans: B

Q29. Which type of anesthesia is preferred for transurethral resection of prostate (TURP)? A. General anesthesia B. Spinal anesthesia C. Epidural anesthesia D. Local anesthesia	Ans: B
Q30. Why is spinal anesthesia preferred for TURP? A. It is cheaper B. Old patients have decreased pulmonary reserve and awake patient can tell central signs of TURP syndrome C. General anesthesia is contraindicated D. It provides better muscle relaxation	Ans: B
Q31. What is the most common irrigating fluid used in TURP? A. Normal saline B. Glycine C. Mannitol D. Sorbitol	Ans: B
Q32. How much fluid is estimated to be absorbed every minute during TURP? A. 5-15 mL B. 10-30 mL C. 30-50 mL D. 50-100 mL	Ans: B
Q33. TURP syndrome is characterized by all of the following EXCEPT: A. Fluid overloading B. Dilutional hyponatremia C. Hypo-osmolality D. Hypernatremia	Ans: D
Q34. Glycine absorbed during TURP can produce: A. Hyperglycemia and hyperammonemia B. Hypoglycemia and hyponatremia C. Hyperkalemia and acidosis D. Hypercalcemia and alkalosis	Ans: A
Q35. Hypo-osmolality and hyponatremia in TURP syndrome can produce: A. Peripheral neuropathy B. Cerebral edema causing restlessness, confusion, or seizures C. Renal failure D. Cardiac arrest	Ans: B

Q36. Hyperglycemia during TURP can produce: A. Hypertension and tachycardia B. Transient visual disturbances including transient blindness C. Permanent blindness D. Hearing loss	Ans: B
Q37. What is the treatment of fluid overload in TURP syndrome? A. Mannitol B. Furosemide C. Dialysis D. Fluid restriction alone	Ans: B
Q38. What is used to treat symptomatic hyponatremia in TURP syndrome? A. Normal saline B. Hypertonic saline C. Ringer lactate D. 5% Dextrose	Ans: B
Q39. What serum potassium level should be corrected before taking the patient for surgery in hyperkalemia? A. < 4.5 mEq/L B. < 5.0 mEq/L C. < 5.5 mEq/L D. < 6.0 mEq/L	Ans: C
Q40. Which of the following is TRUE about anesthetic management of hyperkalemia? A. Ringer lactate is preferred B. Succinylcholine is safe to use C. Potassium-rich fluids should be avoided and succinylcholine is contraindicated D. Hyperventilation should be avoided	Ans: C
Q41. Why is mild hyperventilation beneficial during general anesthesia in hyperkalemic patients? A. It increases oxygen delivery B. It causes alkalosis which pushes potassium intracellularly C. It reduces anesthetic requirement D. It prevents bronchospasm	Ans: B
Q42. What is the general consensus for mild chronic hypokalemia (serum K⁺ >3 mEq/L) before surgery? A. Must be corrected in all cases B. No need to treat if there are no ECG changes or organ dysfunction C. Surgery must be postponed D. Potassium supplementation is mandatory	Ans: B

Q43. In hypokalemic patients on digoxin, what serum potassium level should be achieved before surgery? A. > 3 mEq/L B. > 3.5 mEq/L C. > 4 mEq/L D. > 4.5 mEq/L	Ans: C
Q44. In hypokalemia, the dose of non-depolarizing muscle relaxant should be: A. Increased B. Decreased as prolonged block is expected C. Unchanged D. Doubled	Ans: B
Q45. Why must rapid correction of hyponatremia never be done? A. It causes hyperkalemia B. It can cause pontine myelinolysis C. It causes renal failure D. It causes cardiac arrest	Ans: B
Q46. What is the key principle in anesthetic management of hypercalcemia? A. Avoiding calcium-containing fluids B. Maintenance of hydration and urine output C. Using calcium channel blockers D. Avoiding all muscle relaxants	Ans: B
Q47. Hypocalcemia can produce which of the following during anesthesia? A. Hypertension and tachycardia B. Laryngospasm and hypotension C. Bronchospasm and hyperkalemia D. Malignant hyperthermia	Ans: B
Q48. Hypermagnesemia potentiates the action of: A. Only depolarizing muscle relaxants B. Only non-depolarizing muscle relaxants C. Both depolarizing and non-depolarizing muscle relaxants D. Neither depolarizing nor non-depolarizing muscle relaxants	Ans: C
Q49. Hypomagnesemia may be associated with: A. Hyperkalemia and hypercalcemia B. Refractory hypokalemia and hypocalcemia C. Hypernatremia and metabolic alkalosis D. Polycythemia	Ans: B

Q50. What is the effect of excess fluid administration in renal patients?

- A. It improves renal perfusion
- B. It can precipitate pulmonary edema and fluid deficiency can worsen kidney injury
- C. It has no significant effect
- D. It always improves urine output

Ans: B

Answer Key & Explanations

Q1. Answer: B — Decreased erythropoietin or decreased intake due to anorexia

The anemia in renal impairment is due to decrease erythropoietin or decrease intake due to anorexia seen in uremia.

Topic: Preoperative Evaluation – Anemia | Ref: Ch 22, Page 202

Q2. Answer: D — More than 80%

More than 80% patients with chronic renal disease exhibit hypertension.

Topic: Preoperative Evaluation – Hypertension | Ref: Ch 22, Page 202

Q3. Answer: C — Diabetes

Diabetes is the most common cause of chronic kidney disease as an endocrine abnormality.

Topic: Preoperative Evaluation – Endocrine Abnormalities | Ref: Ch 22, Page 202

Q4. Answer: B — The injury is fully resolved

Patients with acute kidney injury (AKI) should not undergo elective surgery until the injury is fully resolved.

Topic: Timing and Preparation before Surgery | Ref: Ch 22, Page 202

Q5. Answer: B — < 130/80 mm Hg

Control of blood pressure target should be < 130/80 mm Hg in CKD patients before surgery.

Topic: Timing and Preparation before Surgery | Ref: Ch 22, Page 202

Q6. Answer: B — 24 hours

Dialysis should be performed within 24 hours before surgery.

Topic: Timing and Preparation before Surgery | Ref: Ch 22, Page 202

Q7. Answer: C — CNS of these patients is extremely sensitive to sedatives and narcotics

Premedication with benzodiazepines should be avoided as CNS of these patients is extremely sensitive to effect of sedatives and narcotics.

Topic: Premedication | Ref: Ch 22, Page 202

Q8. Answer: B — > 0.5 mL/kg/hr

Aim should be to maintain urine output > 0.5 mL/kg/hr intraoperatively.

Topic: Intraoperative Monitoring – Urine Output | Ref: Ch 22, Page 202

Q9. Answer: B — Fenoldopam

Fenoldopam (selective dopamine-1 agonist) is chosen over dopamine if required for very major surgeries in renal patients.

Topic: Intraoperative Monitoring – Urine Output | Ref: Ch 22, Page 202

Q10. Answer: B — They may be required for hemodialysis in future

If invasive blood pressure monitoring is required then radial and ulnar artery should be spared as they may be required for hemodialysis in future.

Topic: Intraoperative Monitoring | Ref: Ch 22, Page 202-203

Q11. Answer: C — Dorsalis pedis

Dorsalis pedis is most often chosen in these patients for invasive BP monitoring.

Topic: Intraoperative Monitoring | Ref: Ch 22, Page 203

Q12. Answer: B — AV fistula

BP cuff should not be applied in the limb having AV fistula.

Topic: Intraoperative Monitoring | Ref: Ch 22, Page 203

Q13. Answer: B — Associated uremic coagulopathy, uremic polyneuropathy, and intolerance to extra fluids

Central neuraxial blocks should be avoided due to associated uremic coagulopathy, uremic polyneuropathy and intolerance of renal patients to extra fluids to treat hypotension.

Topic: Choice of Anesthesia | Ref: Ch 22, Page 203

Q14. Answer: C — Propofol

Due to shorter half-life and all inactive metabolites, propofol is the induction agent of choice in renal patients.

Topic: General Anesthesia – Induction | Ref: Ch 22, Page 203

Q15. Answer: C — Exaggerated response of CNS due to disrupted blood brain barrier in uremia

Propofol may be required in small doses because of exaggerated response of CNS due to disrupted blood brain barrier in uremia.

Topic: General Anesthesia – Induction | Ref: Ch 22, Page 203

Q16. Answer: C — Desflurane

Desflurane is the inhalational agent of choice because it does not produce fluoride.

Topic: General Anesthesia – Maintenance | Ref: Ch 22, Page 203

Q17. Answer: B — It produces high levels of fluoride approaching renal threshold of 50 $\mu\text{mol/L}$

Sevoflurane should not be used in renal patients as it produces high levels of fluoride approaching renal threshold of 50 $\mu\text{mol/L}$.

Topic: General Anesthesia – Maintenance | Ref: Ch 22, Page 203

Q18. Answer: C — These patients exhibit hyperkalemia

As these patients exhibit hyperkalemia, succinylcholine should not be used.

Topic: Muscle Relaxants | Ref: Ch 22, Page 203

Q19. Answer: B — Atracurium and Cis-atracurium

Due to Hoffman degradation, non-depolarizing muscle relaxants of choice are Atracurium and Cis-atracurium.

Topic: Muscle Relaxants | Ref: Ch 22, Page 203

Q20. Answer: C — Remifentanyl

Remifentanyl is the opioid of choice because it is rapidly metabolized in plasma by esterases.

Topic: Opioids | Ref: Ch 22, Page 203

Q21. Answer: B — Morphine and Pethidine

Long acting like morphine and pethidine are not recommended in renal patients.

Topic: Opioids | Ref: Ch 22, Page 203

Q22. Answer: B — Normal saline causes hyperchloremic metabolic acidosis producing more hyperkalemia

Recent studies have shown that normal saline causes hyperchloremic metabolic acidosis producing more hyperkalemia. Therefore, ringer lactate is preferred over normal saline with close monitoring of potassium levels.

Topic: Selection of Fluids | Ref: Ch 22, Page 203

Q23. Answer: B — Ringer lactate

Ringer lactate is preferred over normal saline with close monitoring of potassium levels.

Topic: Selection of Fluids | Ref: Ch 22, Page 203

Q24. Answer: B — 0.45% saline or 0.18% saline with dextrose

If ringer lactate cannot be used then 0.45% saline or 0.18% saline with dextrose is preferred over normal saline (0.9%).

Topic: Selection of Fluids | Ref: Ch 22, Page 203

Q25. Answer: B — They have been found to exacerbate renal injury

Colloids (hydroxyethyl starch) have been found to exacerbate renal injury.

Topic: Selection of Fluids | Ref: Ch 22, Page 203

Q26. Answer: C — Gelatins (Gelofusine)

If it is mandatory to use a colloid then Gelatins (Gelofusine) is preferred.

Topic: Selection of Fluids | Ref: Ch 22, Page 203

Q27. Answer: C — > 200 kDa

High molecular weight colloids (>MW 200 kDa) must not be used for renal patients.

Topic: Selection of Fluids | Ref: Ch 22, Page 203

Q28. Answer: B — Non-steroidal anti-inflammatory drugs (NSAIDs)

Postoperatively, avoid non-steroidal anti-inflammatory drugs (NSAIDs) for pain management in renal patients.

Topic: Postoperative | Ref: Ch 22, Page 203

Q29. Answer: B — Spinal anesthesia

Spinal anesthesia is preferred over general anesthesia for TURP.

Topic: Anesthesia for TURP | Ref: Ch 22, Page 203

Q30. Answer: B — Old patients have decreased pulmonary reserve and awake patient can tell central signs of TURP syndrome

Spinal anesthesia is preferred because old patients have decreased pulmonary reserve and awake patient can tell central signs of TURP syndrome and bladder perforation.

Topic: Anesthesia for TURP | Ref: Ch 22, Page 203

Q31. Answer: B — Glycine

A large amount of irrigating fluid (most commonly glycine) is used to distend the bladder and wash away blood in TURP.

Topic: TURP Syndrome | Ref: Ch 22, Page 203

Q32. Answer: B — 10-30 mL

It is estimated that 10-30 mL of fluid is absorbed every minute during TURP.

Topic: TURP Syndrome | Ref: Ch 22, Page 203

Q33. Answer: D — Hypernatremia

TURP syndrome produces fluid overloading, dilutional hyponatremia, and hypo-osmolality — not hypernatremia.

Topic: TURP Syndrome | Ref: Ch 22, Page 203

Q34. Answer: A — Hyperglycemia and hyperammonemia

Glycine can also produce hyperglycemia and hyperammonemia.

Topic: TURP Syndrome | Ref: Ch 22, Page 203

Q35. Answer: B — Cerebral edema causing restlessness, confusion, or seizures

Hypo-osmolality and hyponatremia by producing cerebral edema produces restlessness, confusion or even seizures.

Topic: TURP Syndrome – Signs and Symptoms | Ref: Ch 22, Page 203

Q36. Answer: B — Transient visual disturbances including transient blindness

Hyperglycinemia can produce visual disturbances including transient blindness.

Topic: TURP Syndrome – Signs and Symptoms | Ref: Ch 22, Page 203

Q37. Answer: B — Furosemide

Furosemide is used to treat fluid overload in TURP syndrome.

Topic: TURP Syndrome – Treatment | Ref: Ch 22, Page 204

Q38. Answer: B — Hypertonic saline

Hypertonic saline may be required to treat symptomatic hyponatremia and hypo-osmolality.

Topic: TURP Syndrome – Treatment | Ref: Ch 22, Page 204

Q39. Answer: C — < 5.5 mEq/L

Hyperkalemia must be corrected (serum potassium should be < 5.5 mEq/L) before taking the patient for surgery.

Topic: Hyperkalemia | Ref: Ch 22, Page 204

Q40. Answer: C — Potassium-rich fluids should be avoided and succinylcholine is contraindicated

Potassium rich fluids should be avoided and succinylcholine is contraindicated in hyperkalemia.

Topic: Hyperkalemia – Anesthetic Considerations | Ref: Ch 22, Page 204

Q41. Answer: B — It causes alkalosis which pushes potassium intracellularly

Mild hyperventilation is beneficial because hyperventilation causes alkalosis which pushes the potassium intracellularly.

Topic: Hyperkalemia – Anesthetic Considerations | Ref: Ch 22, Page 204

Q42. Answer: B — No need to treat if there are no ECG changes or organ dysfunction

The general consensus is not to treat mild (serum potassium >3 mEq/L) chronic hypokalemia if there are no ECG changes or any organ dysfunction.

Topic: Hypokalemia | Ref: Ch 22, Page 204

Q43. Answer: C — > 4 mEq/L

The exception is patients on digoxin where serum potassium should be >4 mEq/L before taking them for surgery.

Topic: Hypokalemia | Ref: Ch 22, Page 204

Q44. Answer: B — Decreased as prolonged block is expected

Dose of non-depolarizing muscle relaxant is decreased as prolonged block is expected due to muscle weakness in hypokalemia.

Topic: Hypokalemia – Anesthetic Considerations | Ref: Ch 22, Page 204

Q45. Answer: B — It can cause pontine myelinolysis

Rapid correction must never be done; rapid correction can cause pontine myelinolysis.

Topic: Hyponatremia and Hypernatremia | Ref: Ch 22, Page 204

Q46. Answer: B — Maintenance of hydration and urine output

The key principle in hypercalcemia is maintenance of hydration and urine output.

Topic: Hypercalcemia – Anesthetic Considerations | Ref: Ch 22, Page 204

Q47. Answer: B — Laryngospasm and hypotension

Hypocalcemia can produce laryngospasm and hypotension.

Topic: Hypocalcemia – Anesthetic Considerations | Ref: Ch 22, Page 204

Q48. Answer: C — Both depolarizing and non-depolarizing muscle relaxants

Hypermagnesemia potentiates the action of both depolarizing and non-depolarizing muscle relaxants.

Topic: Hypermagnesemia – Anesthetic Considerations | Ref: Ch 22, Page 204

Q49. Answer: B — Refractory hypokalemia and hypocalcemia

Hypomagnesemia may be associated with refractory hypokalemia and hypocalcemia.

Topic: Hypomagnesemia – Anesthetic Considerations | Ref: Ch 22, Page 204

Q50. Answer: B — It can precipitate pulmonary edema and fluid deficiency can worsen kidney injury

Excess fluid can precipitate pulmonary edema and fluid deficiency can worsen kidney injury. Therefore, maintaining fluid balance is essential.

Topic: Intraoperative Monitoring – Fluid Balance | Ref: Ch 22, Page 202